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From Accuracy to Clinical Nuance: A Comparative Study of ICD-10 Coding Models Across Classical, Deep, and LLM Paradigms

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To my beloved family

Abstract

Automatic ICD-10 coding is a major challenge in clinical natural language processing. It enables the structured representation of unstructured clinical narratives for applications such as billing, epidemiology, mortality statistics, and clinical decision making. Recent advances in large language models (LLMs) and Retrieval-Augmented Generation (RAG) systems have improved semantic understanding in medical text processing. However, their practical advantages over classical machine learning and domain-specific transformer architectures remain insufficiently understood, particularly in the context of long-tailed ICD-10 label distributions.

This work conducts a systematic comparative evaluation of four modeling paradigms for automated ICD-10 classification: classical machine learning models (XGBoost and Random Forest), embedding-based neural models (Word2Vec variants), transformer-based architectures (BERT and Bio_ClinicalBERT), and LLM-based architectures (Mistral, Llama, Gemma, and Qwen) with LoRA fine-tuning and Retrieval-Augmented Generation. The evaluated models are compared in multiple dimensions including predictive performance (Accuracy, Micro-F1, and Macro-F1), rare-class sensitivity (Rare-Class F1), hierarchical consistency, calibration error (ECE), and computational efficiency.

Experimental results on a clinical dataset with 20,000 samples and 604 ICD-10 classes identify a fundamental trade-off between overall predictive accuracy and clinical nuance. Transformer-based models attain the highest overall accuracy (up to 0.926) but show near-zero performance on rare diagnostic classes. In contrast, RAG-enhanced LLM systems substantially improve rare-class detection (up to 0.089 Rare-F1) and calibration quality, although at significantly higher computational cost and inference latency. Classical models such as XGBoost demonstrate high computing efficiency, but poor semantic and contextual understanding.

The findings highlight an important limitation of aggregate evaluation metrics in clinical AI systems: high predictive accuracy does not necessarily imply clinical reliability, particularly under long-tailed diagnostic distributions. The study demonstrates that retrieval-augmented LLM architectures improve rare-condition sensitivity and uncertainty calibration, while transformer-based models remain highly competitive for high-frequency coding tasks. Overall, the results support the development of hybrid, cost-aware clinical AI systems that balance predictive performance, interpretability, calibration reliability, and computational feasibility.

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Acronyms

AI Artificial Intelligence.

BCE Binary Cross-Entropy.

BERT Bidirectional Encoder Representations from Transformers.

Bio_ClinicalBERT Biomedical and Clinical Bidirectional Encoder Representations from Transformers.

BoW Bag-of-Words.

ECE Expected Calibration Error.

EHR Electronic Health Record.

F1 Harmonic Mean of Precision and Recall.

FN False Negative.

FNN Feedforward Neural Network.

FP False Positive.

GPU Graphics Processing Unit.

GRU Gated Recurrent Unit.

ICD-10 International Classification of Diseases, Tenth Revision.

ISTAT Istituto Nazionale di Statistica.

LLM Large Language Model.

LoRA Low-Rank Adaptation.

LSTM Long Short-Term Memory.

Macro-F1 Macro-Averaged F1 Score.

Micro-F1 Micro-Averaged F1 Score.

MIMIC Medical Information Mart for Intensive Care.

ML Machine Learning.

MLP Multi-Layer Perceptron.

NLP Natural Language Processing.

PEFT Parameter-Efficient Fine-Tuning.

QLoRA Quantized Low-Rank Adaptation.

RAG Retrieval-Augmented Generation.

ReLU Rectified Linear Unit.

RF Random Forest.

RNN Recurrent Neural Network.

SVM Support Vector Machine.

TF-IDF Term Frequency–Inverse Document Frequency.

TP True Positive.

UN United Nations.

VSM Vector Space Model.

W2V Word2Vec.

WHO World Health Organization.

XGBoost Extreme Gradient Boosting.

Glossary

calibration The agreement between predicted probabilities and observed outcomes.

clinical coding The process of assigning clinical text to standardized diagnostic codes.

grounding Constraining model outputs using external knowledge or trusted sources.

hallucination A generated output that is factually incorrect or unsupported.

long-tailed distribution A distribution in which a small proportion of classes occur frequently while most classes are rare.

multi-label classification A classification setting in which each instance may belong to multiple classes simultaneously.

retrieval The process of selecting relevant external documents or passages based on similarity to an input query.

Chapter 1

Introduction

1.1 Background

The International Classification of Diseases, Tenth Revision (ICD-10) is the global standard for reporting diseases and health conditions . It plays a vital role in epidemiology , public health surveillance , mortality statistics , and clinical decision-making . ICD-10 offers a common coding scheme that allows the systematic aggregation of data and comparisons of morbidity and mortality statistics across institutions and regions[6].

Despite its importance, ICD-10 code assignment from unstructured clinical text especially cause-of-death declarations , is still largely a manual process. It is a time-consuming, resource-intensive task that is subject to inter-coder variability which can result in inconsistencies and delays in reporting[7]. These restrictions have motivated the development of automatic ICD-10 coding systems based on Natural Language Processing (NLP) technologies.

ICD-10 coding is an inherently difficult task as it is a *multi-label* problem. A single clinical record may contain many illnesses related to the immediate, intermediate and underlying causes of mortality. Also, the ICD-10 taxonomy is severely imbalanced with a long-tailed label distribution where a few number of codes are abundant and the bulk are rare. Such mismatch is especially problematic for standard machine learning algorithms which are biased to favor frequent labels at the expense of rare, but clinically relevant illnesses.

Recent advances in deep learning and Large Language Models (LLMs) have led to major improvements in the understanding of clinical writing. Transformer-based architectures such as BERT have achieved great results in text classification tasks, but still have difficulty in reliably capturing unusual diagnostic patterns. Most recently, the integration of LLMs with Retrieval-Augmented Generation (RAG) has demonstrated promise for anchoring predictions in external medical information, potentially enhancing interpretability and clinical relevance [8; 9].

This work explores the shift from *administrative accuracy* (overall predictive performance) to *clinical nuance* (e.g., rare disease detection, hierarchical correctness, and model reliability).

1.2 Research Objectives

The main goal of this work is to offer a comprehensive evaluation of the different modeling paradigms for automated ICD-10 classification, with a particular focus on their ability to capture clinical nuance in long-tailed distributions.

The study seeks to achieve the following objectives:

- **Objective 1: Multi-label Performance Benchmarking in Different Paradigms**
  To establish a systematic comparison between classical machine learning models, transformer-based models, and LLM-based architectures through metrics like Accuracy and Micro-F1, with particular attention on their capacity to handle clinical records with multiple labels.
- **Objective 2: Measuring Clinical Nuance and Long-Tail Performance**
  Comparison of macro-F1 and rare-class F1 across models, particularly for low-frequency ICD-10 codes usually ignored by traditional techniques.
- **Objective 3: Assessing Knowledge Grounding and Hallucination Mitigation**
  To study if Retrieval-Augmented Generation can increase factual accuracy by rooting forecasts on authoritative ICD-10 resources such as the ICD-10 Tabular List and coding rules.
- **Objective 4: Reliability and Calibration Evaluation**
  To analyze the trustworthiness of the model with Expected Calibration Error (ECE) and confidence distributions, and to explore if the predicted probability might contribute to clinical decision making.

1.3 State of the Art: Retrieval-Augmented Generation

Classical machine learning and deep neural network approaches to ICD-10 coding generally struggle with scalability and imbalance of the ICD label space. These approaches can achieve excellent overall accuracy but generally perform poorly in the rare illness categories, which limits their practical utility.

Large Language Models are better at collecting contextual information, modeling complex clinical narratives. However, when used in isolation, they could produce inaccurate or fictitious predictions, a phenomenon generally referred to as *hallucination* [9]. This is especially problematic in clinical applications where reliability and traceability are crucial.

Retrieval-Augmented Generation (RAG), addresses this issue by enhancing LLMs with external data sources. RAG systems in ICD-10 coding acquire important information from organized medical resources, such as the ICD-10 Tabular List and coding rules, and include this information into the production process. This approach enables models to proceed beyond surface-level pattern matching and align predictions with well-established clinical standards.

Recent research suggest that RAG-based systems can enhance both accuracy and interpretability by providing explicit evidence for anticipated codes [8; 9]. In addition, RAG has been found to improve model calibration, resulting to more reliable confidence estimations, which is a crucial need for deployment in clinical workflows.

1.4 Research Contribution

This study makes the following key contributions:

- A systematic empirical comparison of classical, transformer-based, and LLM-based approaches for ICD-10 coding on a large-scale multi-label clinical dataset.
- A multi-dimensional evaluation framework that goes beyond accuracy, and considers rare-class performance, calibration, and hierarchical consistency.
- An analysis of the trade-off between administrative accuracy and clinical nuance, showing that models with lower overall accuracy may perform better on rare and clinically important conditions.
- Investigating the potential of Retrieval-Augmented Generation to improve reliability and mitigate hallucinations in clinical NLP systems.

1.5 Thesis Structure

The remainder of this thesis is structured as follows:

- Chapter 2 describes the theoretical foundation of natural language processing, including word embeddings, transformer architectures, and large language models.
- Chapter 3 provides an overview of existing ICD-10 coding approaches and modeling paradigms.
- Chapter 4 describes the dataset, preprocessing steps, and experimental methodology.
- Chapter 5 describes the experimental results and comparative analysis.
- Chapter 6 discusses the findings, including important trade-offs and implications for clinical deployment.
- Chapter 7 concludes the thesis and provides directions for future work.

Chapter 2

The Fundamentals

2.1 Introduction

This chapter describes the theoretical basis of automatic ICD-10 coding from clinical literature. It covers the main ideas behind classical machine learning, natural language processing (NLP), neural networks, transformer structures, large language models (LLMs), retrieval-augmented generation, and parameter-efficient fine-tuning. Particular focus is given to multi-label classification, long-tailed label distributions and probabilistic calibration, which are the main issues in clinical coding systems.

2.2 International Classification of Diseases

The International Classification of Diseases (ICD) is a systematic classification of diseases, causes of mortality and health conditions. It dates back to 1893. Since 1946 it has been maintained and periodically amended by the World Health Organization (WHO) [10]. The most recent edition, ICD-11, was published in 2022, while ICD-10, published in 1993, is widely used in research and practice. Its continued prevalence is largely due to the availability of historical data encoded in ICD-10, including the dataset used in this study.

ICD-10 is a standardized and hierarchical classification system that allows for the consistent recording, reporting and analysis of health data over time and countries. The categorization is arranged in 22 chapters, each one dealing with a broad category of disorders or health-related situations, generally classified by organ systems or domains, as defined in the official instruction manual [1]. The chapters are recognized by the first letter of the ICD code.

Within each chapter there are blocks that further group related conditions at a more granular level. Blocks are a range of alphanumeric codes. Categories are subsequently broken into subcategories of increasing clinical specificity. This hierarchy is reflected in the composition of ICD-10 codes, which are normally three to five characters long. The first three characters of the codes indicate the category, then a decimal point and extra characters are used to specify subcategories. This hierarchical approach allows for uniform encoding of clinical information and allows for large-scale statistical analysis. Figure 2.1 shows the hierarchical structure of ICD-10 codes , which consist of chapters, blocks, categories and subcategories.

[Chapter I]

A00 - B99 Certain infectious and parasitic diseases

↪ [Block]

A00 - A09 Intestinal infectious diseases

↪ [Category]

A00 Cholera

↪ [Subcategory]

A00.0 Cholera due to Vibrio cholerae 01, biovar cholerae

A00.1 Cholera due to Vibrio cholerae 01, biovar eltor

A00.9 Cholera, unspecified

Figure 2.1. Sample structure of ICD-10: from chapter to subcategory [1]

The ICD (International Classification of Diseases) is one of the cornerstones of epidemiology, the study of the distribution and determinants of health-related states in specific populations. The ICD standardizes the classification of diseases and makes it possible to produce comparable mortality and morbidity statistics. It is also useful for disease surveillance and the identification of health patterns in time and space [11]. These capabilities are essential for understanding population health dynamics, including disease emergence, population-level risk factors, and long-term health trends.

From the point of view of official statistics, the ICD-coded data are the backbone of the national and international statistical systems. Statistical agencies employ ICD data to compile key indicators such as cause-specific mortality, life expectancy, and illness burden. These indicators are crucial for evidence-based policymaking, resource allocation, and health system planning [12]. The use of ICD in official statistics promotes uniformity, comparability and reproducibility, and hence meets international statistical standards.

The assignment of ICD codes from unstructured clinical narratives such as death certificates and discharge reports is still a complex and resource consuming process, albeit standardized. It is subject to change and requires specialist knowledge, which can alter the quality and dependability of official data [7]. This difficulty is exacerbated in large-scale systems when textual data are in abundance, varied and complex.

ICD coding presents several machine learning challenges, particularly multi-label prediction, hierarchical label dependencies, and severe class imbalance. First, it is a multi-label classification problem by nature, since several codes can be assigned to a document simultaneously. Second, the hierarchical structure of ICD induces dependencies among labels, which models must learn across different levels of granularity. Third, the label distribution is highly imbalanced and long-tailed, with rare but clinically important diagnoses underrepresented [13]. Finally, clinical narratives are often long and noisy, requiring models to identify sparse diagnostic signals within large amounts of text.

These challenges reveal the limitations of purely parametric models that rely only on learned representations and may be less effective for rare conditions or evolving medical knowledge. Retrieval-Augmented Generation (RAG) offers a principled way to incorporate external knowledge into the modeling process in this context. RAG combines dense retrieval and generative modeling, allowing systems to dynamically access relevant clinical information and improve both accuracy and robustness [5].

In particular, RAG is suited for ICD coding in the context of official statistics where uniformity, traceability and interpretability are of utmost importance. RAG addresses important limitations of large language models by conditioning predictions on retrieved evidence, offering greater transparency and reducing reliance on implicit model knowledge, making it a promising approach for scalable and reliable coding systems in large-scale health data environments.

2.3 Machine Learning for Text Classification

2.3.1 Classical Models

Classical machine learning models for text classification are based on features manually extracted from the input text. Typical representations include Bag-of-Words (BoW), n-grams and TF-IDF, which convert unstructured text into numerical vectors suitable for processing by machine learning models [14]. While these techniques are computationally efficient and perform well in a number of tasks, they are highly dependent on manual feature engineering and domain knowledge, which limits transferability across datasets and languages.

Support Vector Machines (SVMs), Naïve Bayes and Decision Trees are common classical models used for text classification. SVMs generate separation hyperplanes in high-dimensional feature spaces and often work well for sparse representations [15]. Decision Trees provide interpretable rule-based structures, while Naïve Bayes models facilitate efficient probabilistic categorization based on simplifying independence assumptions. These models have been typical tools for high-dimensional text categorization for decades. But their emphasis on global feature engineering often hinders them from catching local syntactic nuance in clinical narratives.

Ensemble approaches such as Random Forest combine several decision trees trained on bootstrap samples. They are known to reduce variance and enhance resilience [16]. More sophisticated algorithms like Extreme Gradient Boosting (XGBoost) broaden this paradigm by enhancing the prediction performance by gradient boosting with a second-order approximation of the loss function [17].

Although effective, classical models have serious limitations. They operate on sparse surface-level representations and hence neglect semantic linkages and contextual reliance in text. Furthermore, their effectiveness is heavily dependent on feature engineering. Which limits their generalization capability to complicated linguistic patterns, especially in clinical writing where meaning depends strongly on context.

Such constraint inspired the development of deep learning systems that learn representations automatically from data, in a hierarchical and semantically meaningful way. Compared to traditional models [18], neural architectures require less manual feature engineering and have better ability to capture the complex structure of natural language.

2.3.2 Multi-Label Classification and Long-Tail Distributions

ICD-10 coding is essentially a *multi-label classification* challenge, in which same clinical record might be assigned numerous diagnosis codes. This is unlike standard multi-class classification, where labels must be disjoint and models must account for dependencies across disease states [19]. Formally, the output space is defined as:

$$y \in \{0,1\}^K$$

where K denotes the number of possible labels.

Training in this setting is typically performed using Binary Cross-Entropy (BCE), applied independently to each label:

$$\mathcal{L}_{BCE} = -\sum_{i=1}^{K} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

A major problem in ICD-10 classification is severe class imbalance, where the label distribution is long-tailed or Zipfian [13]. In such distributions, a small number of codes appears very frequently while the majority appear rarely. This imbalance is especially pronounced in large-scale clinical datasets, where rare diagnoses may have only a few training examples. Prior work on structured label spaces has shown that standard classification methods often struggle with long-tailed ICD code distributions, and that zero-shot or few-shot techniques may be needed to associate rare clinical descriptions with their corresponding codes [20].

This imbalance may be seen in datasets such as MIMIC-III, where the majority of occurrences are attributed to a narrow distribution of high-frequency codes, whereas a large number of codes are sparsely populated [13]. Thus, typical models tend to prefer common classes, resulting in high overall accuracy at the expense of rare but clinically significant illnesses.

This leads to a known problem in medical coding called the *accuracy paradox*, where aggregate performance measurements can disguise poor performance on low-frequency labels that are typically the most relevant for clinical decision-making. Hence, a systematic study of precision, recall and F-score is needed to appropriately assess model utility in these imbalanced scenarios [21]. Clinical applications also require credible estimates of uncertainty, in addition to discrete measures . Deep neural networks are prone to overconfidence in their predictions and so probabilistic calibration, which aligns the expected probability with the actual likelihood, is crucial for trustworthy ICD-10 coding [22]. In automatic mortality coding, a well-calibrated model that produces a 90% confidence score for a cause of death should be correct around 90% of the time.

Moreover, clinical narratives are usually extensive, loud and varied with useful and irrelevant information. This is a *needle-in-a-haystack* challenge where models need to find sparse diagnostic signals in enormous volumes of text. This makes studying unusual classes even harder.

These issues motivate the use of advanced architectures like transformer-based models and retrieval-augmented techniques that are more suited for learning robust representations in long-tailed clinical scenarios.

2.4 Natural Language Processing (NLP)

Natural Language Processing (NLP) allows machines to understand and process human language [23]. In the clinical sector Natural Language Processing (NLP) is defined as the conversion of unstructured medical narratives (e.g. discharge summaries, death certificates) into a structured format that may be utilized for categorization, information extraction and decision support [24].

Clinical NLP differs from general domain NLP in domain-specific terminology, acronyms, irregular formatting, and highly specialized language patterns. These characteristics pose challenges to representation learning and necessitate the use of contextual and domain-adapted language models for health care applications.

2.4.1 Text Representation and the Vector Space Model

Standard NLP approaches use the *Vector Space Model* (VSM) to represent documents as vectors in a high-dimensional space $\mathbb{R}^V$, where V is the vocabulary size. This change from basic counting of frequencies to representing meaning in vector spaces is a major step in computational linguistics, driving the discipline towards richer document semantics [25]. The model assesses the similarity between texts by their relative position in this space . However the model is limited by the sparseness of the vocabulary dimension .

- **Bag-of-Words (BoW):** BoW depicts documents as sparse vectors of term frequencies and ignores the order of words and semantic relationships. It is simple and effective for text categorization, but it suffers from vocabulary mismatch and cannot capture contextual meaning [14]. This is particular problem when the same semantic concepts are expressed by a number of different phrases, or when the sense of the word is context-dependent. BoW only captures shallow semantics and cannot model deeper semantic linkages.

- **TF-IDF Weighting:** TF-IDF is an extension of BoW with term specificity. The weight $W_{i,j}$ for the i-th phrase in the j-th document is as follows:

$$W_{i,j} = \mathrm{tf}_{i,j} \cdot \log\left(\frac{N}{\mathrm{df}_i}\right)$$

where N is the total number of documents and df_i is document frequency of term i [26]. TF-IDF is an improvement of BoW by giving more weight to discriminative terms and less weight to common words. However, it still does not capture semantic linkages, word order and contextual dependencies [14]. This leaves TF-IDF still limited in its ability to capture nuanced clinical text, therefore motivating the development of more advanced techniques such as word embeddings and transformer-based models [27].

2.4.2 Word Embeddings and Distributed Semantics

Word embeddings are based on the *Distributional Hypothesis* that asserts that words that appear in comparable settings tend to have similar meanings. This notion, defined by Zellig Harris in 1954 [28], is the linguistic basis of modern distributional semantics, which states that lexical meaning is a function of a word's surroundings rather than its isolated definition [25; 27]. This linguistic principle is operationalized in big corpora using word embeddings which transform the principle into dense numerical representations by mapping semantic proximity to geometric distance.

Word2Vec implements this idea by learning dense vector representations that maximize the average log probability of context words [29; 30]. This optimization maps semantically related words to neighboring points in a continuous embedding space.

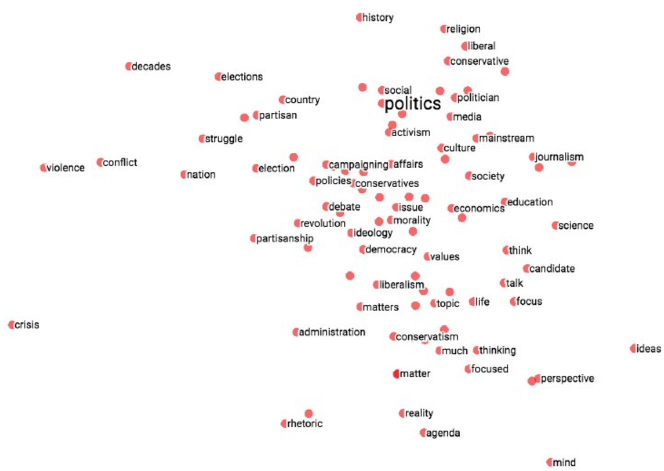

Figure 2.2. Sample of semantic relationships in word embeddings, where vector arithmetic captures linguistic regularities (adapted from [2]).

A key characteristic of this framework is that meaningful linear correlations between word vectors develop and linguistic regularities can be articulated by vector arithmetic [2]. For example, semantic analogies of the form:

$$king - man + woman \approx queen$$

demonstrate that embedding spaces encode syntactic and semantic structure in a mathematically consistent manner.

Formally, the training objective can be expressed as:

$$\mathcal{L} = \frac{1}{T} \sum_{t=1}^{T} \sum_{\substack{-c \leq j \leq c \\ j \neq 0}} \log P(w_{t+j} \mid w_t)$$

This representation allows models to learn latent semantic structure and generalize better than sparse representations such as TF-IDF . However, Word2Vec embeddings are *context-independent*, i.e., a word is represented by a single vector regardless of its usage. This constraint is particularly relevant in clinical writing, where phrases may have context-dependent interpretations, motivating contextual embedding models.

2.4.3 Contextual Embeddings

Contextual embeddings solve the limits of static representations by dynamically producing word vectors conditioned on surrounding context. This paradigm change is possible largely by the Transformer architecture introduced in *Attention Is All You Need* [4] that replaces recurrence and convolution with attention mechanisms alone.

At the heart of this architecture is the *self-attention* mechanism, allowing each token in a sequence to consider all other tokens simultaneously, therefore modeling long-range dependencies and contextual interactions.

Furthermore, self-attention may be calculated in parallel and can directly reflect global interactions between tokens, which is not possible for earlier models such as RNNs and LSTMs which process sequences sequentially and struggle with long-range dependencies. The result is more effective training and a large increase in representational power.

This architecture is the basis of models such as BERT which apply bidirectional self-attention to produce contextualized embeddings in which the representation of a word is dependent on both its left and right context [31]. This is particularly true in clinical NLP, where medical terminology can have a very context-dependent meaning and often depend on the surrounding diagnostic information for their meaning. Bio_ClinicalBERT and other domain-specific transformer versions go a step further to tailor these contextual representations to biomedical and clinical corpora, resulting in improved performance on healthcare-related NLP tasks.

As a result, transformer-based contextual embeddings have become the backbone of modern NLP systems achieving considerable advancements in tasks such as text categorisation, entity recognition and clinical coding.

2.5 Neural Networks for Text Modeling

2.5.1 Feedforward Neural Networks

Feedforward Neural Networks Feedforward neural networks (FNNs) are among the simplest neural network architectures in machine learning, consisting of a sequence of linear transformations followed by nonlinear activation functions. Structurally, these networks form a directed acyclic graph, with information flowing only from the input layer to the output layer, without any feedback links [18].

A typical FNN consists of an input layer that takes in feature vectors, one or more hidden layers that learn intermediate representations, and an output layer that produces the prediction. Each neuron computes a weighted sum of its inputs and applies a non-linear activation function:

$$y = \sigma(Wx + b)$$

where W and b are learnable parameters and $\sigma(\cdot)$ is a non-linear activation function like ReLU or sigmoid.

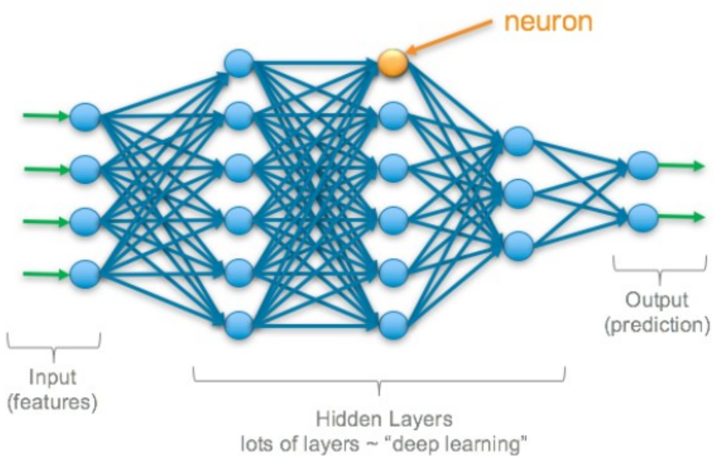

Figure 2.3. Deep learning neural architecture [3]

FNNs are useful for problems with static input representations, such as classification and regression, due to the absence of cyclic connections, which allows steady forward propagation. Deep networks with numerous hidden layers are able to learn the hierarchical feature representations and eventually translate the raw input data to the higher level abstractions [18].

For classification problems, the output layer is usually modeled with a softmax activation function to provide class probabilities. However, feedforward networks are not suitable for modeling sequential or temporal dependencies, since they do not have mechanisms for modeling context across input position. This limitation fueled the creation of recurrent architectures such as RNN and LSTM, but also attention-based architectures.

Feedforward neural networks, notably multi-layer perceptrons (MLPs), are still a fundamental building block of deep learning and remains widely utilized thanks to their simplicity, computational economy and good performance on structured data tasks.

2.5.2 Recurrent Neural Networks and Gradient Stability

Recurrent neural networks (RNNs) are intended to model sequential data through a hidden state that changes over time. The concealed state at time step t is updated by:

$$h_t = \sigma(W_{hh}h_{t-1} + W_{xh}x_t + b)$$

where h_t represents the hidden state, x_t the input, and W_{hh} and W_{xh} are learnable weight matrices.

RNNs are theoretically capable of learning long-range dependencies but training is typically hampered by the difficulties of *vanishing* and *exploding gradients* [32; 33]. Backpropagation through time (BPTT) propagates gradients across several time steps, causing the repeated multiplication of Jacobian matrices. This method can be expressed as:

$$\frac{\partial \mathcal{L}}{\partial h_t} \propto \prod_{k=1}^{t} W_{hh}$$

When the spectral radius of W_{hh} is less than one, the gradients decay exponentially (vanishing gradients) and the network finds it hard to learn long-term dependencies. If it is greater than one, the gradients erupt and the numerical instability happens.

These problems greatly restrict the capability of standard RNNs to memorize information from prior time steps, especially for long sequences. This entails that RNNs are less capable at capturing long-range dependencies, and this is insufficient for context-aware understanding over longer text spans such as clinical narratives.

To overcome these limitations, complex architectures like Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs) use gating techniques to regulate the flow of information and preserve the stability of the gradient over time [34]. These architectures help improve learning of long-range relationships and prior to the advent of transformer-based models dominated sequence modeling jobs.

2.5.3 Long Short-Term Memory (LSTM)

To overcome the vanishing gradient problem in typical RNNs [34], Long Short-Term Memory (LSTM) networks were proposed. Unlike traditional recurrent designs, LSTMs use a memory cell and a collection of gates to govern the flow of information over time.

An LSTM cell has a hidden state h_t and a cell state C_t . The cell state is a long-term memory and can store information for many time steps. There are three main gates that control the flow of information:

- **Forget Gate:** Determines which information from the previous cell state should be discarded:
$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$
- **Input Gate:** Controls which new information is added to the cell state:
$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$
- **Candidate State:** Generates new candidate values to be incorporated:
$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

The cell state is then updated as:

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$

Finally, the output gate determines the hidden state:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o), \quad h_t = o_t \cdot \tanh(C_t)$$

Intuition. The key innovation of LSTMs is the cell state update equation. The additive structure provides a *constant error pathway* that permits gradients to flow through time steps substantially unmodified when $f_t \approx 1$. This method prevents the exponential decay of the gradient and helps the network to remember long-term dependencies.

Relevance to Clinical Text. Critical diagnostic information in clinical narratives may surface early in the document, but influence coding decisions later. LSTMs can remember such information over extended sequences and are more appropriate for structured medical text than vanilla RNNs .

Limitations. LSTMs work well, but they need to handle data sequentially, which reduces the ability to parallelize and increases training time. They also find it difficult to capture very long-range dependencies as compared to attention-based models. This constraint caused the switch to transformer architectures.

2.6 Transformer Architectures

Transformer architectures are a paradigm shift in sequence modelling introduced by *Attention Is All You Need* [4]. Transformers do not process tokens sequentially, but use self-attention processes exclusively. It allows parallel computation and greatly enhances efficiency and performance..

The key element of the Transformer is the *self-attention mechanism*. This mechanism enables each token in a sequence to attend to all other tokens, thus capturing global dependencies irrespective of their distance. The mechanism is described as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

where Q, K, and V represent the query, key, and value matrices derived from the input sequence.

A major advantage of this formulation is that it does not reoccur. All tokens can be processed in parallel, which allows rapid training on modern hardware and better modeling of long-range dependencies than RNN-based designs.

To further improve the representational power, transformers employ *multi-head attention*, enabling the model to attend to different portions of the input sequence simultaneously. Each attention head learns various relationships. This allows the model to capture syntactic and semantic patterns in different subspaces of the representations.

Because self-attention does not encode word order, transformers employ *positional encoding* to insert sequence information into the model. This respects token order and parallel compute.

In addition, architectural elements like residual connections and layer normalization help stabilize training and enable the construction of deeper networks.

In this architecture, models like BERT use bidirectional self-attention to generate contextualized representations, and have achieved state-of-the-art results on many NLP tasks [31].

Modern large language models are built on top of transformer architectures that enable scale contextual representation learning through self-attention. These developments directly enabled the emergence of instruction-tuned LLMs, and retrieval-augmented systems for domain-specific applications such as clinical coding.

Transformers are now the dominating paradigm in NLP. They are the backbone of huge language models and have enabled substantial progress on tasks requiring contextual awareness and long-range dependency modeling.

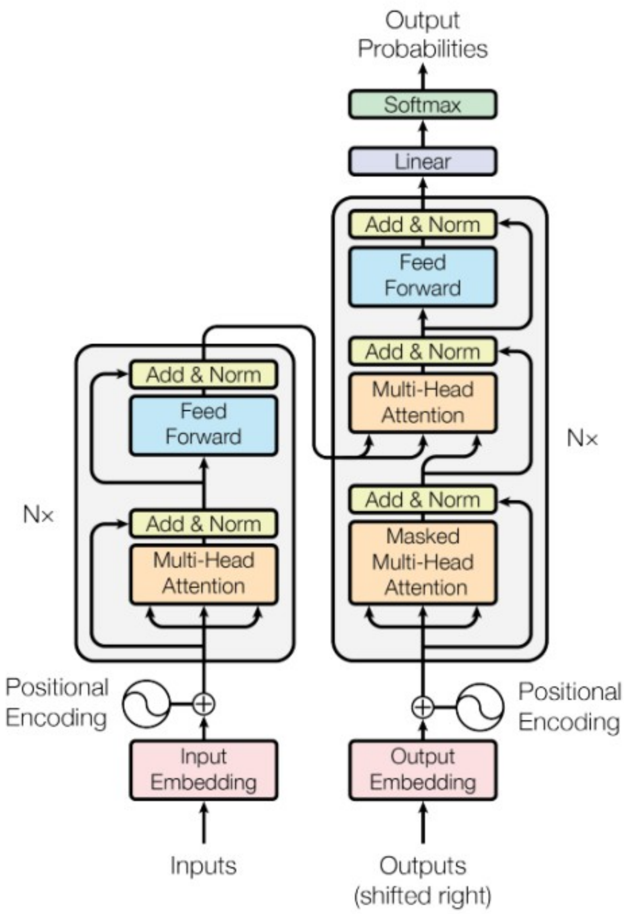

Figure 2.4. The original Transformer model architecture [4]

2.7 Large Language Models and Retrieval-Augmented Generation

Large Language Models (LLMs) are transformer-based architectures trained on huge datasets to execute a wide variety of natural language tasks. These models are further fine-tuned by reinforcement learning from human feedback (RLHF) to align model outputs better with user instructions and downstream tasks [35]. However, LLMs do have great generalization capabilities, but they are based largely on *parametric knowledge*, which is set at the conclusion of the training period. This may lead them to generate obsolete, incomplete or factually wrong information, which is commonly referred to as *hallucination* and is a serious issue in automated medical coding.

To address these issues, RAG (Retrieval-Augmented Generation) incorporates external knowledge sources into the generation process [5]. Instead of depending solely on its internal parameters, RAG enhances the input with pertinent information retrieved from a knowledge base, allowing the model to produce more precise and contextually appropriate outputs.

The RAG framework consists of three core steps:

- **Retrieval:** For an input query, a retriever module locates relevant documents from an external corpus via a similarity search in a dense vector space.
- **Augmentation:** The retrieved documents are combined with the original input, giving the model additional context.
- **Generation:** The language model takes the augmented input and produces the final output.

This procedure can be viewed as supplying LLMs with an external, dynamic memory that lets them to access current and domain-specific information during inference.

Retrieval is generally dependent on similarity measurements, such as cosine similarity, formally:

$$\mathrm{sim}(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \|\mathbf{d}\|}$$

where $\mathbf{q}$ is the query embedding and $\mathbf{d}$ is a document embedding.

Conceptually, this works by conditioning generation on retrieved evidence, which decreases hallucinations and enhances factual consistency . Recent evaluations of LLMs in healthcare indicate that such grounding is not just a performance boost but a critical safety necessity for clinical validation within hospital settings [36].

In actual deployments, dense retrieval is often expedited by vector indexing libraries like FAISS (Facebook AI Similarity Search) which allows efficient nearest-neighbor retrieval over massive collections of embeddings.

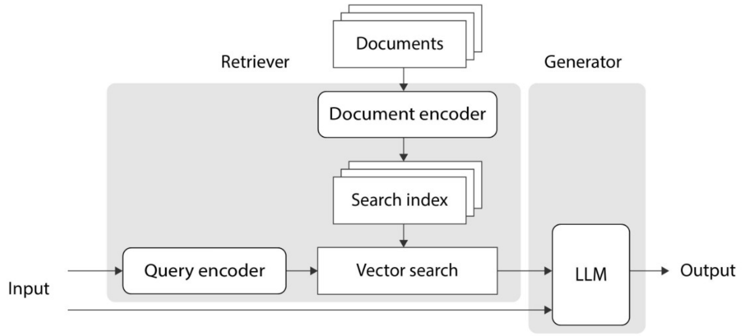

Figure 2.5. RAG architecture showing the interaction between the retriever and generator (adapted from [5]).

Traditional fine-tuning embeds knowledge directly into the model parameters, while RAG stores knowledge apart from the model, which is trained without it, leading to a more flexible approach. This allows systems to be modifiable on-the-fly without retraining the entire model.

In summary, Retrieval-Augmented Generation is a major step towards combining the generative prowess of LLMs with the reliability of external knowledge sources, paving the way for more trustworthy and interpretable AI systems.

2.8 Parameter-Efficient Fine-Tuning (PEFT)

Finetuning all the parameters of large language models (LLMs) is computationally expensive and is normally not feasible for large models. Parameter-Efficient Fine-Tuning (PEFT) addresses this issue by tuning only a small set of parameters while keeping most of the pretrained weights fixed and significantly reduces the memory and compute requirements.

A popular PEFT approach is *Low-Rank Adaptation (LoRA)* [37], which factorizes weight updates into low-rank matrices. Instead of directly updating a pretrained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA expresses the update as:

$$W' = W_0 + \Delta W, \quad \Delta W = BA$$

where:

- $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$
- $r \ll \min(d, k)$ is the rank of the decomposition

This reparameterization reduces the number of trainable parameters from $O(dk)$ to $O(r(d + k))$, allowing for efficient fine-tuning even for very large models.

In reality, LoRA injects these low-rank matrices into certain components inside transformer designs, such as the attention projection layers, e.g., the query and value matrices. During training, the original weights are frozen and only the low-rank matrices are modified.

$$h = (W_0 + BA)x$$

Intuition. The efficacy of LoRA is based on the *low intrinsic rank hypothesis* which states that task-specific adaptations to large pretrained models reside in a low-dimensional subspace. LoRA constrains the updates to a low-rank structure that captures the most relevant directions of change, while avoiding to change the whole parameter field.

Comparison with Other PEFT Methods. Alternative PEFT approaches include:

- **Adapters:** These add extra trainable modules inside the network, freezing the pretrained weights, and so increase the depth of the model without changing the original parameters [38].
- **BitFit:** A sparse fine-tuning approach where only the bias terms of the model are updated, resulting in a very lightweight adaptation strategy for specific downstream tasks [39].

These approaches are also economical in training, but LoRA has the benefit of an efficient-performance trade-off, typically providing performance close to full fine-tuning, especially in low- and medium-resource environments.

Relevance to This Work. In the scope of ICD-10 coding, LoRA permits big language models, such as Mistral and LLaMA, to be adapted easily, avoiding the extremely expensive process of comprehensive fine-tuning. This is especially relevant in the context of the magnitude of clinical data and the requirement to evaluate different topologies with limited resources.

Overall, PEFT approaches such as LoRA offer a scalable and practical means to deploy big language models in real-world clinical settings, striking a balance between performance, efficiency, and adaptability.

2.9 Evaluation Metrics and Calibration

To evaluate ICD-10 categorization, metrics must address the multi-label nature of ICD-10 codes and the substantial class imbalance. In such scenarios, conventional accuracy is not enough as it can be dominated by frequent classes. Rather, accuracy, recall and f1-score are metrics that are often used to quantify performance.

The precision and recall for a given class i are defined as:

$$\text{Precision}_i = \frac{TP_i}{TP_i + FP_i}, \quad \text{Recall}_i = \frac{TP_i}{TP_i + FN_i}$$

The F1-score is the balance between precision and recall and is computed as:

$$F1_i = \frac{2 \cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}$$

2.9.1 Micro and Macro Averaging

For multi-label classification, F1 scores are averaged over classes using different averaging algorithms.

Micro-F1. Micro-F1 combines the true positives, false positives and false negatives for all classes:

$$\text{Micro-F1} = \frac{2 \cdot \sum_i TP_i}{2 \cdot \sum_i TP_i + \sum_i FP_i + \sum_i FN_i}$$

This measure is dominated by labels with many instances and so focuses on frequent classes. Therefore, Micro-F1 often captures the overall performance in imbalanced datasets.

Macro-F1. Macro-F1 is the unweighted mean of per-class F1 scores:

$$\text{Macro-F1} = \frac{1}{K} \sum_{i=1}^{K} F1_i$$

This treats all classes equally, being more sensitive to performance on rare labels. Hence, Macro-F1 provides a more balanced assessment of model performance in long-tailed distributions.

In clinical coding, the difference between Micro-F1 and Macro-F1 is important, as uncommon diagnoses might be clinically relevant even though they are infrequent..

2.9.2 Calibration and Reliability

In addition to prediction performance, the trustworthiness of model confidence estimations needs to be evaluated as well. When a model is well calibrated, the probability estimate it gives is the true probability of occurrence.

Expected Calibration Error (ECE) evaluates the difference between the projected confidence and the true accuracy by dividing the predictions into M bins [22]:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left| \text{acc}(B_m) - \text{conf}(B_m) \right|$$

where:

- B_m is the set of predictions in bin m
- $\text{acc}(B_m)$ is the empirical accuracy of bin m
- $\text{conf}(B_m)$ is the average predicted confidence

A lower ECE means greater calibration, i.e. the projected probabilities are closer to the actual results. This is especially critical in clinical situations, where overconfident but wrong predictions might undermine trustworthiness and thus negatively affect decision-support dependability.

2.10 Summary

This chapter addressed the theory behind automatic ICD-10 coding systems, including classical machine learning, neural architectures, transformer models, retrieval-augmented generation, and parameter-efficient fine-tuning methods. Special attention was placed on the issues of multi-label classification, long-tailed label distributions, contextual language interpretation and probabilistic calibration in clinical NLP.

These concepts are the theoretical foundation of the experimental methodology provided in the next chapter, where the discussed architectures are implemented and assessed using real-world ICD-10 mortality data.

Chapter 3

ICD-10 Coding Models

3.1 Problem Formulation

Automatic ICD-10 coding can be formulated as a multi-label classification problem where each clinical document can be labeled with numerous diagnosis codes simultaneously. The label space is technically defined as:

$$y \in \{0,1\}^K$$

where K denotes the total number of ICD-10 classes and each y_i indicates the presence or absence of a specific code.

Given an input document x, the model produces logits $z \in \mathbb{R}^K$, which are transformed into probabilities using a sigmoid activation:

$$\hat{y}_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}}$$

Training is performed using Binary Cross-Entropy (BCE), applied independently to each label:

$$\mathcal{L}_{BCE} = -\sum_{i=1}^{K} \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right]$$

This formulation implies conditional independence of the labels, simplifying the optimization but not directly modeling connections between co-occurring disorders. Such dependencies may be crucial in the therapeutic setting as some conditions tend to co-occur.

3.1.1 Long-Tailed Label Distribution

The ICD-10 coding has a unique long-tailed label distribution, where a few codes appear frequently, but the majority are infrequent [13]. Let $p(k)$ be the frequency of class k. Empirically this distribution is a power-law:

$$p(k) \propto k^{-\alpha}$$

for some $\alpha > 0$.

This imbalance creates a number of problems. First, the BCE objective is biased towards frequent classes and thus biased optimization. Second, the gradient does not sufficiently update the infrequent classes, which limits the generalization. Third, models tend to overfit to common patterns and under-represent clinically important but unusual diseases.

Hence, this imbalance has to be rectified for the creation of clinically relevant ICD-10 coding systems.

3.1.2 Label Dependency Modeling

The Binary Cross-Entropy (BCE) formulation implies conditional independence between labels, and treats each ICD-10 code as an independent binary prediction. This makes the optimization easy but overlooks the structured links between medical problems.

In practice ICD codes tend to be connected. Often, cardiovascular illness occurs in association with respiratory or renal disease. This assumption of independence is comparable to:

$$P(y_1, y_2, \ldots, y_K \mid x) = \prod_{i=1}^{K} P(y_i \mid x)$$

which does not hold for clinical data when relationships among diagnosis are widespread.

This constraint can lead to inconsistent predictions, such as selecting mutually exclusive diagnoses, or missing clinically relevant co-occurrence patterns.

The challenge has led to a variety of approaches. In graph-based models, ICD codes are considered nodes in the hierarchy or co-occurrence graph, allowing dependencies to be explicitly modelled. For example, hierarchical classification techniques use the structure of ICD taxonomies to enforce consistency between parent and child codes. Recently, neural techniques have proposed label attention mechanism and graph neural network to capture the inter-label interactions.

However, modeling label relationships remains a challenge, particularly in large-scale situations with thousands of potential codes. We assume independence here for scalability and comparability among models, but acknowledge the limits as a significant direction for further work .

3.2 Classical Machine Learning Models

3.2.1 Representation Learning with Word2Vec

Classical models rely on static representations extracted from the Word2Vec embeddings [29]. Each word w_i is associated with a dense vector $\mathbf{v}_{w_i} \in \mathbb{R}^d$ that is learned by maximizing the probability of context words under the distributional hypothesis.

Mean pooling is used to generate document-level representations:

$$\mathbf{v}_{doc} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{v}_{w_i}$$

This approach provides a computationally inexpensive approximation of semantic information, but does not take into account the order of words and contextual connections, limiting its expressiveness.

3.2.2 Random Forest

Random Forest models are an ensemble of decision trees trained using bootstrap samples of the data. Each tree recursively splits the feature space and predictions are aggregated by averaging or voting.

Formally, the model can be expressed as:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} f_t(x)$$

where f_t represents a single decision tree. Random sampling of samples and features can reduce the variation and increase the generalization.

However, the application of static input features constrains the capacity of Random Forest models to learn intricate language patterns.

3.2.3 XGBoost

XGBoost is an extension of gradient boosting, that builds an additive model of decision trees optimized using second-order gradient information [17]:

$$\hat{y} = \sum_{k=1}^{T} f_k(x)$$

At each iteration a new tree is fitted to minimize a regularized objective:

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

where Ω is a penalty for model complexity.

We combine the benefits of XGBoost on structured features with the expressiveness of the input embeddings, which are kept limited in their representational power.

3.3 Sequential Deep Learning Models

3.3.1 Word2Vec + LSTM

To leverage sequential information, Word2Vec embeddings are used as input to a Long Short Term Memory (LSTM) network [34]. The hidden state at time step t is computed as:

$$h_t = \text{LSTM}(x_t, h_{t-1})$$

The final hidden state h_T is used for classification and encodes the document representation.

LSTMs solve the problems with static embeddings by learning temporal dependencies and context over sequences. The gating mechanisms enable selective remembering and forgetting of information . This helps with the vanishing gradient issue.

However, their sequential structure restricts the potential for parallelization and scalability, especially for long clinical papers.

3.4 Transformer-Based Models

3.4.1 Self-Attention Mechanism

Transformer models do not use recurrence but self-attention, which lets each token attend to all the other tokens in the sequence. The attention operation is formulated as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

This method allows for the direct modeling of long-range dependencies and lends itself to parallel processing

3.4.2 BERT-based Classification

BERT encodes the input text into contextual embedding by utilizing stacked self-attention layers [31]. Classification is done on the special [CLS] token representation:

$$\hat{y} = \sigma(Wh_{CLS} + b)$$

BERT's bidirectional encoding enables it to learn rich semantic associations, which is suitable for text categorization tasks.

3.4.3 Bio_ClinicalBERT

Bio_ClinicalBERT is an extended BERT model that further pre-trains on clinical [40]. This adaption models the domain-specific terminology more accurately, however the benefits are domain dependent, depending on how well the pretraining data aligns with downstream tasks.

3.5 Large Language Models and Fine-Tuning

Large Language Models (LLMs) are transformer architectures augmented with billions of parameters that allow them to describe complex linguistic and semantic structures [41]. These models can be instruction-tuned or fine-tuned for classification tasks.

3.5.1 Parameter-Efficient Fine-Tuning (LoRA)

Fine-tuning LLMs in full is computationally expensive. Low-Rank Adaptation (LoRA) [37] tackles this by factorizing weight updates to low-rank matrices:

$$W' = W_0 + \Delta W, \quad \Delta W = BA$$

where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with $r \ll \min(d, k)$.

The forward pass becomes:

$$h = (W_0 + BA)x$$

This formulation decreases the number of trainable parameters greatly while maintaining model expressiveness. The low intrinsic rank hypothesis provides an explanation for the success of LoRA, which assumes task-specific adaptations are in a low-dimensional subspace.

3.6 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation integrates external information into the prediction process to overcome the constraints of parametric knowledge [5]. Given a query q, relevant documents d are found via similarity:

$$\mathrm{sim}(q, d) = \frac{q \cdot d}{\|q\| \|d\|}$$

The retrieved context is concatenated to the input and passed through the language model to enable grounded predictions.

By providing access to structured medical knowledge in ICD-10 coding, RAG leads to improved performance on rare and confusing diagnoses and hallucination reduction.

3.7 Unified Modeling Perspective

The models explored in this work can be viewed as a sequence in representational capacity:

- Classical models rely on static embeddings and shallow decision boundaries
- Sequential models introduce temporal context through recurrence
- Transformers enable global context via attention
- LLMs extend this with large-scale pretraining and semantic reasoning
- RAG augments models with external knowledge

Each stage addresses limitations of the previous one, particularly in handling long-range dependencies and rare classes.

3.8 Summary

This chapter presented theoretical background for the ICD-10 classification modeling approaches. It models the problem as multi-label learning under long-tailed distributions and demonstrates the difficulties for different model classes.

The shift from classical models to retrieval-augmented LLMs suggests an expanded capacity to model semantic complexity and rare clinical occurrences. In the next chapter we discuss the experimental setup utilized to test these models under comparable and controlled conditions .

Chapter 4

Experimental Setup

4.1 Dataset

This study uses a dataset from the Italian National Institute of Statistics (ISTAT) with around 600,000 death records for the period 2022-2023 all over Italy. The records contain written clinical descriptions tied to ICD-10 diagnosis codes for causes of death.

To make the computation feasible, a representative subset of 20,000 samples is picked that preserves the statistical properties of the original dataset. This subset size balances computational limitations with statistical representativeness given restricted GPU resources. The preprocessed dataset comprises 604 unique ICD-10 classes.

Downsampling does not remove the fundamental properties that make real-world clinical data challenging: extreme class imbalance and a long-tailed label distribution in which a few diagnoses are abundant and others are rare [13]. This imbalance poses huge hurdles for multi-label classification algorithms, especially in predicting unusual clinical diseases.

Figure 4.1 presents the distribution of the 15 most frequent ICD-10 classes in the processed dataset. The image demonstrates the extreme skewness of the label distribution with just a few diagnostic categories covering the dataset and the remainder of the ICD-10 codes being weakly represented. This long-tailed behavior promotes the use of assessment metrics such as Macro-F1 and Rare-Class F1 to better assess model performance on rare diagnoses.

4.2 Preprocessing

A standard pipeline is applied to all samples to transform raw clinical narrative text into structured representations that may be used by machine learning. Text normalization (lowercasing and noise removal), tokenization, ICD-10 code extraction using regular expressions, and label transformation into multi-hot encoded vectors

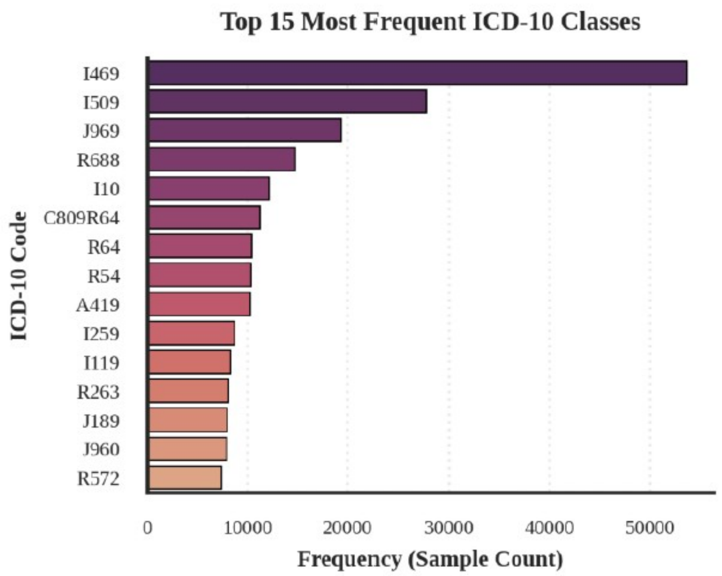

Figure 4.1. Distribution of the 15 most frequent ICD-10 classes in the dataset

Table 4.1. Structure of the Raw ICD-10 Dataset (Pre-Translation)

Field	Description	Example
`id`	Unique identifier for each record	491704
`line_number`	Position of the diagnosis within the death certificate (0 = primary cause)	0
`cause_code`	ICD-10 code(s), potentially containing multiple labels and temporal annotations	I509(3Months); J969(3Months)
`original_text`	Raw clinical description in Italian	SCOMPENSO CARDIOCIRCOLATORIO CON INSUFF RESPIRATORIA
`cause_interval`	Interval between onset and death (if available)	03M

for multi-label classification.

The preparation pipeline further normalizes ICD-10 annotations by extracting standardized ICD-10 codes from multi-label entries and removes embedded temporal expressions from code strings (e.g. "3Months"). Code normalization guarantees consistency in the annotations . However in the case of multi-label categorization , several labels are still maintained .

To better fit English-domain pre-trained transformer and LLM structures, clinical descriptions are translated from Italian to English using Google Translate.

Then, model-specific tokenization algorithms are implemented for transformer and LLM-based architectures to ensure alignment with their respective vocabularies and input constraints.

For the original dataset format with noisy and semi-structured annotations with multi-label ICD-10 codes and associated temporal metadata, see Table 4.2. The processed result after normalization, code standardization and translation for comparison is shown in Table 4.3. This transformation is necessary to allow transformer and LLM-based models to generalize well on clinical narratives without being biased by formatting issues and non-clinical artifacts.

Table 4.2. Raw Clinical Records Before Preprocessing

Cause Code (Raw)	Original Text (Italian)
I469	ARRESTO CARDIACO
I509(3Months); J969(3Months)	SCOMPENSO CARDIOCIRCOLATORIO CON INSUFFICIENZA RESPIRATORIA
I469	ARRESTO CARDIOCIRCOLATORIO
I480(14Years); N189; I351	FIBRILLAZIONE ATRIALE PAROSSISTICA 14A; INSUFFICIENZA RENALE CRONICA; INSUFFICIENZA VALVOLA AORTICA
R570	COLLASSO CARDIO RESPIRATORIO

4.3 Train–Validation–Test Split

The data is separated into training (70%), validation (15%) and test (15%) subsets. This split guarantees enough data for model learning and independent sets for hyper-parameters tuning and unbiased evaluation. The validation set is utilized for threshold tuning and parameter selection. The test set is used for final performance evaluation.

Due to the substantial imbalance of clinical coding data, we consider unusual classes as ICD-10 labels with less than five occurrences in the training dataset. This definition allows targeted investigation of low frequency diagnoses and reflects the long-tailed distribution often encountered in clinical coding jobs.

Table 4.3. Processed Dataset After Cleaning and Translation

Clean Text (Italian)	Code	Clean Text (English)
arresto cardiaco	I469	cardiac arrest
scompenso cardiocircolatorio con insufficienza respiratoria	I509	cardiovascular failure with respiratory failure
arresto cardiocircolatorio	I469	cardiovascular arrest
fibrillazione atriale parossistica insufficienza renale cronica insufficienza valvola aortica	I480	paroxysmal atrial fibrillation with chronic renal failure
collasso cardio respiratorio	R570	cardiorespiratory collapse

4.4 Hyperparameters and Model Configuration

In order to allow a fair comparison between paradigms all experiments are performed in standardized and controlled training conditions. Since the tests are conducted with a constrained computing budget on Google Colab NVIDIA L4 GPU settings, the hyperparameters are chosen to satisfy a trade-off between predictive performance, computational feasibility, and hardware limits.

4.4.1 Classical Machine Learning Models

Classical ML models use Word2Vec document embeddings created via mean pooling of token embeddings. As baselines we choose efficient and resilient methods like XGBoost and Random Forest for comparison with respect to neural architectures.

Table 4.4. Hyperparameters for Classical Machine Learning Models

Model	Configuration
Word2Vec + Random Forest	200 decision trees using bootstrapped sampling and feature randomness.
Word2Vec + XGBoost	300 trees; maximum depth = 6; learning rate = 0.1.

4.4.2 Deep Learning Models

We use LSTM networks on top of Word2Vec embeddings for implementing sequential neural architectures. The hidden size and the dropout rate are adjusted to balance the representational capacity and the control of overfitting.

Table 4.5. Hyperparameters for Deep Learning Models

Parameter	Value
Hidden size	128
Dropout	0.3
Batch size	8
Learning rate	1×10^{-3}
Epochs	3
Maximum sequence length	128

4.4.3 Transformer-Based Models

To keep the uniformity for the studies with BERT-base and Bio_ClinicalBERT, the optimization settings of the fine-tuning of the transformer topologies are kept the same.

Table 4.6. Hyperparameters for Transformer Models

Parameter	Value
Learning rate	2×10^{-5}
Batch size	16
Epochs	3
Maximum sequence length	64
Gradient accumulation	2

We group LLM-based studies into two categories: 1) normal QLoRA fine-tuned language models without retrieval augmentation, and 2) retrieval-augmented (RAG) variants that retrieve external ICD-10 knowledge during inference. This provides a direct comparison between pure LLM reasoning and retrieval-augmented generating methodologies.

4.4.4 Large Language Models and LoRA Fine-Tuning

All LLM experiments use Parameter-Efficient Fine-Tuning (PEFT) with QLoRA and 4-bit quantization [42] because complete fine-tuning is computationally too expensive. This considerably reduces GPU memory utilization while keeping competitive performance.

4.4.5 Detailed LLM Architecture Parameters

In addition to the training hyperparameters, the large language models we analyze differ greatly in their architectural scale and number of parameters. Table 4.8 shows

Table 4.7. Hyperparameters for LLM Fine-Tuning

Model	Configuration
Mistral-7B-Instruct-v0.3	QLoRA fine-tuning; 4-bit quantization; LoRA rank = 8; learning rate = 2×10^{-4}; gradient accumulation = 4; maximum sequence length = 128.
Llama-3.2-3B-Instruct	QLoRA fine-tuning; 4-bit quantization; LoRA rank = 8; learning rate = 2×10^{-4}; gradient accumulation = 4; maximum sequence length = 128.
Gemma-3-4B-it	QLoRA fine-tuning; 4-bit quantization; LoRA rank = 8; learning rate = 2×10^{-4}; gradient accumulation = 2; maximum sequence length = 128.
Qwen2.5-3B-Instruct	QLoRA fine-tuning; 4-bit quantization; LoRA rank = 8; learning rate = 2×10^{-4}; gradient accumulation = 2; maximum sequence length = 128.

the total number of parameters, trainable LoRA parameters and the proportion of trainable parameters for each model.

Table 4.8. Architectural and Fine-Tuning Parameters of LLM-Based Models

Model	Model Size	Total Parameters	Trainable Parameters	Trainable %
Mistral-7B-Instruct-v0.3	7B	3.64B	13.0M	0.36%
Llama-3.2-3B-Instruct	3B	1.81B	7.9M	0.44%
Gemma-3-4B-it	4B	2.50B	7.3M	0.29%
Qwen2.5-3B-Instruct	3B	1.70B	4.6M	0.27%

4.5 Inference Setup

All models are assessed on the held-out test set with the best checkpoints selected on the validation set. For traditional and deep learning models, predictions are made by immediately applying the trained classifier to each input document. For multi-label prediction, sigmoid outputs are thresholded to binary labels with thresholds tuned on the validation set.

For transformer based models the [CLS] representation is passed to a classification head producing a logit for each class of ICD-10. Predicted probabilities are calculated by:

$$\hat{y}_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}}$$

where z_i is the logit corresponding to class i. The estimated probabilities for each class are thresholded separately to generate the final set of labels.

Inference for RAG-augmented LLM models is done in a Retrieval-Augmented Generation (RAG) framework. We develop an external knowledge base of ICD-10 by leveraging publicly available reference datasets of ICD-10 from Kaggle and the official ICD-10 documentation from the World Health Organization (WHO). The knowledge base includes organized mappings between ICD-10 codes, diagnostic descriptions and accompanying clinical terminology.

At inference time, the input clinical narrative is embedded and used as a query to retrieve semantically relevant ICD-10 items from the knowledge base utilizing FAISS (Facebook AI Similarity Search) for efficient vector similarity retrieval. The retrieved context is appended to the original clinical text and then fed to the language model to provide grounded predictions.

This retrieval approach provides the model with access to external medical knowledge at inference time, thus enhancing factual consistency, allowing the model to recognize rare disorders. The retrieval augmentation is only done during inference time, and not during parameter fine-tuning.

To be compatible with the multi-label classification framework, we confine the generated outputs to the ICD-10 label space used in the assessment process. The FAISS retrieval index uses vector representations from the `all-MiniLM-L6-v2` SentenceTransformer embedding model.

To improve robustness, the inference is performed with the same tokenization, max sequence length and retrieval settings as used during validation. This guarantees the given findings are all under consistent test-time conditions.

Inference is done in batches for efficiency and no gradient modifications are performed during assessment.

4.5.1 External Knowledge Base

The retrieval part of the RAG process uses an external knowledge base of ICD-10 built from publicly available medical coding resources. The key reference resources are publicly available ICD-10 datasets on Kaggle [43] and the official WHO ICD-10 documentation [1]. The resources consist of structured relations between diagnostic codes, disease descriptions and medical terminology for retrieval-augmented inference.

This resulting knowledge base provides an external memory layer that allows LLM-based systems to retrieve clinically relevant contextual information beyond their pre-trained parametric knowledge.

Table 4.9. External Knowledge Sources Used in the RAG Framework

Source	Type	Purpose
ISTAT Mortality Records	Clinical Dataset	Training and evaluation of ICD-10 coding models.
Kaggle ICD-10 Dataset	Knowledge Base	Retrieval augmentation for LLM-based models.
WHO ICD-10 Documentation	Reference Standard	Validation of ICD-10 descriptions and hierarchy mapping.

4.6 Evaluated Models

To provide a complete comparison across modeling paradigms, we examine classical machine learning models, a deep learning baseline, transformer-based architectures, and big language models with and without retrieval augmentation . Classical machine learning and LSTM-based models work with Word2Vec embeddings but transformer- and LLM-based architectures are fine-tuned on clinical text directly. Retrieval-augmented variants further inject external ICD-10 knowledge at inference stage via RAG framework.

The full list of models tested during the experiments is shown in Table 4.10.

Table 4.10. Evaluated Models

Category	Models
Classical Machine Learning	Word2Vec + Random Forest; Word2Vec + XGBoost
Deep Learning	Word2Vec + LSTM
Transformer-Based Models	BERT-base; Bio_ClinicalBERT
LLM-Based Models (QLoRA)	Llama-3.2-3B-Instruct; Gemma-3-4B-it; Qwen2.5-3B-Instruct; Mistral-7B-Instruct-v0.3
RAG-Enhanced LLM Models	RAG-Llama-3.2-3B-Instruct; RAG-Gemma-3-4B-it; RAG-Qwen2.5-3B-Instruct; RAG-Mistral-7B-Instruct-v0.3

This experimental design enables direct comparison between feature-based classifiers, sequential neural architectures, transformer encoders, and instruction-tuned LLMs, while also isolating the impact of retrieval augmentation within the LLM setting.

4.7 Hardware and Training Time

All experiments are performed on Google Colab's NVIDIA L4 GPUs. The experimental setting is conducted using an approximate computational budget of €100, which is a realistic limit that is often encountered in applied academic research.

Training time varies a lot between modeling paradigms. Classical models like XGBoost may be trained in just a few minutes, but transformer-based models like BERT take over 40 minutes. Training retrieval-augmented LLMs takes significantly longer. Mistral-RAG takes over 625 minutes to train.

Experimental results show a trade-off between computational efficiency and representational power. Classical models are efficient but do not have a good knowledge of context. In contrast, LLM-based systems obtain more semantic contextual information at a much higher computing cost.

4.7.1 Software Environment

All experiments are implemented in Python because of its rich ecosystem for machine learning and natural language processing. We use Scikit-learn and XGBoost to create classical machine learning models. For deep learning and transformer based architectures we use PyTorch and the Hugging Face Transformers package.

Large language models are fine-tuned in a parameter-efficient manner with the PEFT and QLoRA frameworks. Data preprocessing and analysis is performed using Pandas and NumPy. Model evaluation and visualization are conducted with Matplotlib, Seaborn and Scikit-learn packages.

The experiments are performed on Google Colab with the NVIDIA L4 GPUs.

4.8 Evaluation Protocol

All models are evaluated in the same experimental settings for fair comparison. We use the same test set for all experiments, and prediction thresholds are tuned on the validation set. For multi-label predictions, sigmoid activations are applied individually to each output class.

Performance is measured in terms of Exact Match Accuracy, Micro-F1, Macro-F1, Top-3 Accuracy, Top-5 Accuracy, Rare-Class F1 and Hierarchical Accuracy. Together, these measures assess general predictive accuracy and sensitivity to low-frequency diagnoses.

This is especially crucial in our setting, as the label distribution is highly uneven and Macro-F1 considers the frequent and the rare classes as equally important. Exact Match Accuracy is also added as a stringent metric, which requires the predicted label set of each sample to be exactly the same as the ground truth labels.

Table 4.11. Software Frameworks and Libraries

Tool / Framework	Purpose
Python	Primary programming language used for implementation.
Scikit-learn	Classical machine learning models and evaluation utilities.
XGBoost	Gradient boosting framework for ensemble learning.
PyTorch	Deep learning framework for neural architectures.
Hugging Face Transformers	Implementation and fine-tuning of transformer and LLM-based models.
PEFT / QLoRA	Parameter-efficient fine-tuning of large language models.
FAISS	Vector similarity search for retrieval-augmented generation.
Pandas / NumPy	Data preprocessing and numerical computation.
Matplotlib / Seaborn	Visualization and experimental plotting.
Google Colab	Cloud-based GPU training environment.

4.8.1 Calibration

Besides the predictive performance, we also quantify the model dependability with Expected Calibration Error (ECE) which evaluates the agreement between projected probabilities and actual outcomes [22]. Reliable confidence estimation is critical in clinical decision support settings, where miscalibrated predictions may erode trust and interpretability.

4.9 Reproducibility

All models are trained and evaluated on the same dataset splits, using the same preprocessing pipelines and evaluation techniques to allow comparability and reproducibility. To increase reproducibility of training and evaluation results, we fix random seeds across experiments. These controls ensure that variation in observed performance is due to model architecture and not experimental variation.

4.10 Summary

This chapter defined the experimental approach used to test all models under controlled settings. The study standardizes the preprocessing pipelines, hyper-parameter setups, evaluation protocols, and hardware constraints to enable a fair comparison of traditional machine learning, deep learning, transformer-based, and LLM-based systems. In the next chapter we show the results of quantitative and qualitative evaluation acquired in this framework.

Chapter 5

Evaluation and Results

5.1 Overview

This chapter gives a detailed evaluation of the suggested ICD-10 coding models in terms of prediction performance, sensitivity to rare classes, calibration reliability and computing efficiency. Trade-offs between aggregate prediction accuracy and clinically significant performance for long-tailed ICD-10 label distributions are discussed in detail.

We evaluate traditional machine learning methods, deep learning architectures, transformer-based models and retrieval-augmented large language models (LLMs). Our results indicate that transformer-based architectures yield the best overall predictive performance and that retrieval-augmented systems improve greatly the detection of rare classes and probabilistic reliability.

5.2 Overall Performance

Table 5.1 shows the performance of models on the main evaluation metrics, including Exact Match Accuracy, Micro-F1, Macro-F1, Top-3 Accuracy, Top-5 Accuracy, and Rare-F1.

Table 5.1. Overall Performance of ICD-10 Coding Models

Model	**Accuracy**	**Micro-F1**	**Macro-F1**	**Top-3**	**Top-5**	**Rare-F1**
BERT-base	**0.927**	**0.948**	0.352	0.981	0.984	0.0023
Bio_ClinicalBERT	0.924	0.944	0.362	0.982	0.984	0.0045
XGBoost (W2V)	0.893	0.931	**0.383**	0.954	0.959	–
Mistral-7B (RAG)	0.873	0.913	0.356	0.978	0.981	0.0846
LLaMA-3.2 (RAG)	0.858	0.912	0.366	0.981	0.982	**0.0890**
Gemma-3 (RAG)	0.838	0.902	0.361	0.976	0.980	0.0890
Qwen2.5 (RAG)	0.795	0.889	0.358	0.980	0.983	0.0876

Transformer-based models achieve the highest overall predictive performance, particularly in terms of Exact Match Accuracy and Micro-F1. BERT-base performs best overall with 92.7% accuracy and 0.948 Micro-F1 score. Bio_ClinicalBERT performs similarly well, demonstrating that general-domain pretraining is still quite competitive in specialized clinical categorization scenarios.

However, the comparatively smaller variation in Macro-F1 scores across architectures suggests that improvements in aggregate performance are driven primarily by frequent ICD-10 categories rather than balanced performance across the full label space.

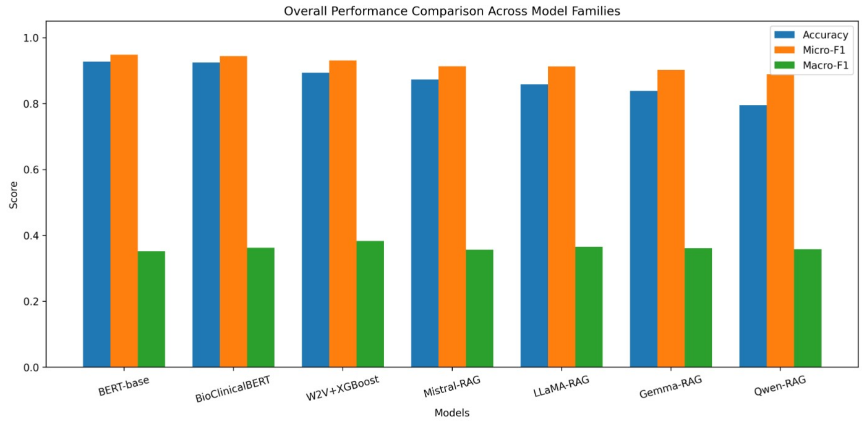

Figure 5.1. Comparison of overall predictive performance among model families

The overall performance is shown in Figure 5.1, where transformer-based models dominate on aggregate evaluation metrics, but retrieval-augmented LLMs remain competitive on Macro-F1 performance, albeit with lower overall accuracy. This indicates that LLM-based systems generalize better to varied clinical narratives rather than optimize mainly for dominant labels.

Interestingly, the Word2Vec + XGBoost baseline achieves the highest Macro-F1 score (0.383), indicating rather balanced performance over frequent and infrequent classes. This shows that simpler statistical learning algorithms can still be competitive in highly imbalanced label distributions.

Overall, the results show that high aggregate accuracy does not necessarily correlate with good rare-class sensitivity or balanced clinical reasoning skill.

5.3 Domain Adaptation Analysis

5.3.1 BERT vs Bio_ClinicalBERT

We compare BERT-base with Bio_ClinicalBERT to evaluate the effect of domain-specific pretraining.

Surprisingly, BERT-base marginally exceeds Bio_ClinicalBERT in terms of overall accuracy (92.7% vs 92.4%). These findings indicate that large scale pre-training on general domain data can compensate for the absence of specialized clinical corpora, particularly for large multi-label classification problems that require different contextual representations.

Bio_ClinicalBERT shows marginal improvements in Macro-F1 and Rare-F1 . The relatively tiny difference shows that pretraining in the clinical domain alone may not be enough to overcome the problems of long-tailed ICD-10 distributions.

5.3.2 Comparison with Retrieval-Augmented LLMs

LLM-based systems (Mistral, LLaMA, Gemma, and Qwen) show lower aggregate accuracy but are more sensitive to unusual diagnoses and complex clinical narratives . This shows that retrieval augmentation is more important than domain-specific pretraining alone for improving rare-class performance.

> *Domain-specific pretraining alone cannot ensure the superiority in large-scale multi-label clinical classification tasks.*

The results also imply that external retrieval grounding may be more significant than specialized pretraining for handling low-frequency clinical circumstances.

5.4 Long-Tail Analysis

5.4.1 Rare-Class Performance

We evaluate the performance on rare-class with Rare-F1, i.e. on low-frequency ICD-10 codes.

Table 5.2. Rare-Class Performance Comparison

Model	Rare-F1	Improvement vs BERT
BERT-base	0.0023	1×
Bio_ClinicalBERT	0.0045	2×
Mistral-7B (RAG)	0.0846	37×
LLaMA-3.2 (RAG)	**0.0890**	**39**×
Gemma-3 (RAG)	0.0890	39×
Qwen2.5 (RAG)	0.0876	38×

Table 5.2 shows that there are substantial differences in rare-class sensitivity across model families. Retrieval-augmented LLMs obtain about 40× improvement in Rare-F1 compared to transformer baselines, while preserving competitiveness in overall classification performance.

RAG-LLaMA and RAG-Gemma produce the best results for rare-class performance, with Rare-F1 scores close to 0.089. Transformer-based models, on the other hand, achieve the best aggregate accuracy but have near-zero rare-class sensitivity.

These results provide empirical support for a major hypothesis of this work: optimizing for aggregate accuracy alone can result in poor performance on clinically relevant but rare diagnoses.

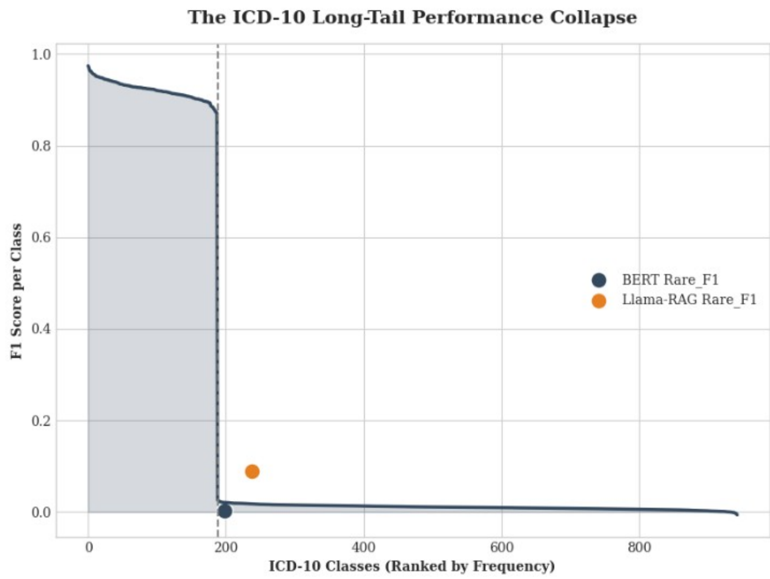

Figure 5.2. Performance degradation across long-tailed ICD-10 label distributions

Figure 5.2 illustrates a severe performance degradation for low-frequency ICD-10 labels. Transformer models achieve better performance on prevalent categories but suffer a large performance gap on rare conditions, which reveals the weakness of accuracy optimization in long-tailed distributions.

Figures 5.2 and 5.3 together show that it is possible to get a high aggregate accuracy with very low sensitivity for the rare classes.

> *High overall accuracy is mostly influenced by common diagnoses and obscures poor performance on rare but clinically important diseases.*
>
> *This result underlines the shortcomings of traditional evaluation measures for clinical AI systems.*

This is one of the main conclusions of this study, and it highlights the need of evaluating clinical AI systems beyond typical aggregate metrics.

5.5 Parameter Efficiency and Computational Trade-offs

5.5.1 Training Efficiency

Model efficiency is evaluated by training time and predicted performance per unit time.

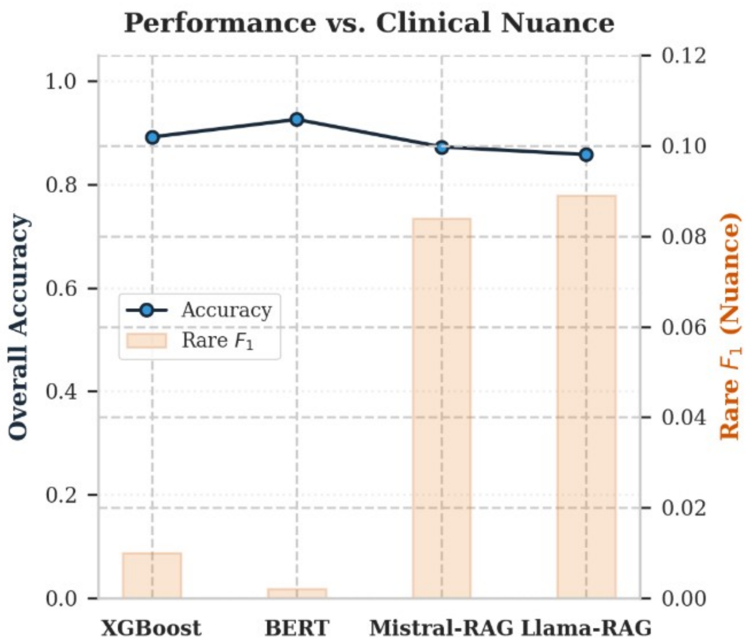

Figure 5.3. The accuracy trap: high aggregate accuracy masking poor rare-class detection

Table 5.3. Training Efficiency Across Models

Model	Training Time (min)	Micro-F1	F1/min
XGBoost	5.42	0.931	**0.172**
BERT-base	43.39	0.948	0.022
Mistral-7B (RAG)	625.02	0.913	0.00146
LLaMA-3.2 (RAG)	298.29	0.912	0.00306
Gemma-3 (RAG)	258.62	0.902	0.00349
Qwen2.5 (RAG)	220.02	0.889	0.00404

Table 5.3 shows significant differences in computational efficiency across different model families. Classic machine learning algorithms such as XGBoost give you with very strong efficiency with competitive prediction performance.

In contrast, retrieval-augmented LLMs demand significantly more processing resources despite increased sensitivity and calibration performance for rare classes.

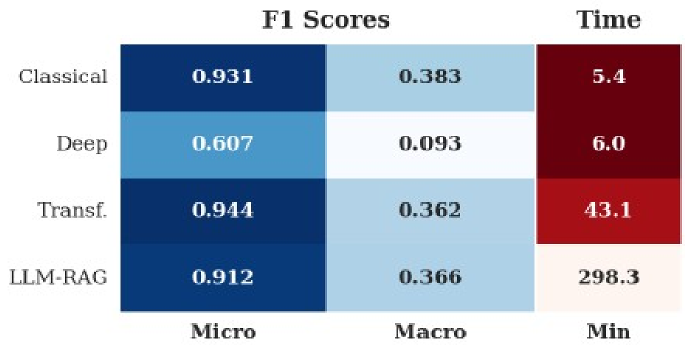

Figure 5.4. Trade-off between computational cost and rare-class performance across model architectures

Figure 5.4 illustrates the trade-off between computational efficiency and clinical nuance. Classical techniques give fast training and great aggregate performance while retrieval-augmented LLMs obtain better rare-condition sensitivity with much higher computational cost.

5.5.2 LoRA Fine-Tuning

LoRA is a parameter-efficient fine-tuning technique that allows to adapt big language models while training only <1% of the full model parameters [37]. This significantly lowers the memory footprint, computing power, and time compared to complete fine-tuning and enables experimentation on consumer-grade cloud GPUs.

Although still computationally expensive compared to classical approaches, LoRA-based systems provide a viable balance between scalability and predictive performance in resource-constrained contexts.

5.5.3 Discussion

> *There is an obvious trade off between computational efficiency and clinical delicacy.*

Classical machine learning algorithms are still quite efficient but have poor semantic reasoning skills . Retrieval-augmented LLMs show remarkable improvements in terms of contextual knowledge and rare-class sensitivity, but demand much larger computational resources during both training and inference.

5.6 Calibration and Reliability

5.6.1 Expected Calibration Error (ECE)

Calibration analysis shows that the confidence reliability varies a lot among model families.

Table 5.4. Calibration Performance Using Expected Calibration Error (ECE)

Model	ECE Before	ECE After
BERT-base	0.00170	0.00040
Bio_ClinicalBERT	0.00162	0.00042
Mistral-7B (RAG)	0.00046	0.00027
LLaMA-3.2 (RAG)	0.00047	0.00027
Gemma-3 (RAG)	0.00041	0.00032
Qwen2.5 (RAG)	0.00056	0.00030

As shown in Table 5.4, retrieval-augmented models exhibit the smallest Expected Calibration Error, which means their predicted confidence scores are better aligned with the true outcomes.

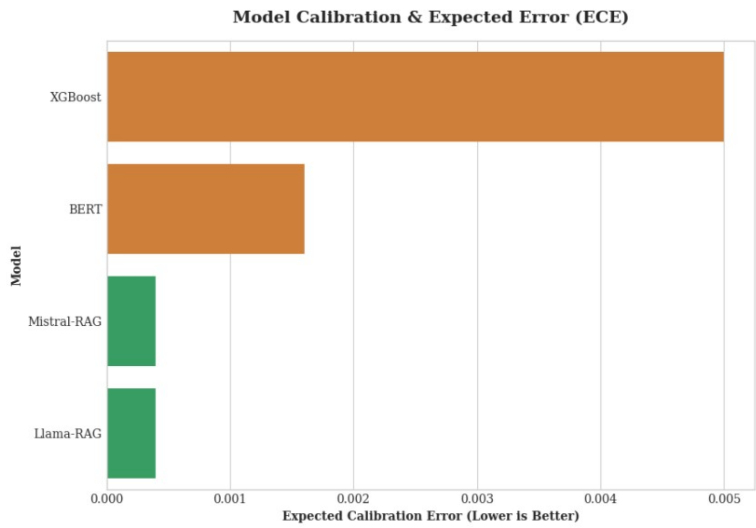

Figure 5.5. Expected Calibration Error (ECE) before and after temperature scaling

We also observe in Figure 5.5 that retrieval-augmented architectures always exhibit less calibration error for all models investigated. This means that retrieval grounding increases not just the prediction quality but also the confidence reliability .

5.6.2 Clinical Implications

Reliable calibration is especially relevant in clinical decision-support systems, because confidence estimations may influence subsequent medical interpretation and human decision-making.

The enhanced calibration of retrieval-augmented systems shows that such models are not only more informative, but potentially more trustworthy for high-stakes settings such as clinical decision making.

RAG-based models provide both improved rare-class sensitivity and more reliable confidence estimates.

5.7 Qualitative Analysis

The qualitative evaluation reveals significant behavioral variations across model families. Retrieval-augmented LLMs show a stronger ability to interpret complex clinical narratives and multi-step diagnostic linkages.

Transformers tend to over-predict common ICD-10 categories, while classical machine learning models fail to generalize beyond superficial lexical patterns. Without retrieval augmentation, LLMs sometimes generate unsupported or hallucinated codes, pointing out the importance of external grounding processes.

Representative prediction examples highlight the different behaviors of classical, transformer-based, and retrieval-augmented LLM systems, especially in contextual reasoning, rare-class sensitivity and hallucination mitigation.

5.8 Summary of Findings

The experiments produced four main results.

First, great aggregate accuracy does not mean that the performance is clinically meaningful in the long-tailed distributions of ICD-10.

Second, retrieval-augmented LLMs provide a substantial boost in sensitivity to rare clinical diseases, with almost $40\times$ improvement in Rare-F1 over transformer baselines.

Third, retrieval grounding significantly increases calibration reliability and confidence estimate.

Finally, there is a clear trade-off between computational efficiency and clinical nuance.

Figure 5.6 summarizes the behavior of the retrieval-augmented LLM framework, showing stable optimization and improved sensitivity to clinically complex cases.

Overall, the results suggest that future clinical coding systems could benefit

Figure 5.6. Training dynamics and evaluation performance of the LLAMA-RAG framework

from hybrid designs that combine the efficiency of transformer classifiers with the contextual grounding of retrieval-augmented language models.

Chapter 6

Discussion

6.1 Overview

In this chapter, we review the experimental results from Chapter 5 with a focus on providing explanations for the model behavior and discussing implications for real-world clinical deployment. The results reveal an underlying and persistent tension between predictive accuracy, clinical nuance, computational efficiency, and reliability. Rather than pinpointing a single best model, the analysis demonstrates that different modelling paradigms optimise for different aspects of the problem.

6.2 Why BERT Achieves the Highest Accuracy

The transformer based models, especially BERT, consistently achieve the greatest overall accuracy and Micro-F1 scores. This is mainly because they learn rich contextual representations from large-scale pre-training and efficiently adapt to structured classification problems via fine-tuning [31].

However, the apparent superiority should be taken with caution. The distribution of ICD-10 is very skewed, with few diagnoses being very frequent in the dataset. Therefore, models that focus on these dominating patterns can become highly performing globally at the expense of rare but clinically relevant illnesses.

> *BERT excels in learning dominant statistical patterns, but struggles to capture the intricacies of unusual clinical events.*

This constraint is not a failure of the model architecture but rather a mismatch between optimization targets and therapeutic considerations.

6.3 Why LLMs Excel in Rare Disease Detection

In contrast, LLM-based models, especially those with retrieval-enabled models, show a great improvement on unusual ICD codes. The performance boost in Rare-F1 (up to $\sim 40\times$ vs. BERT) suggests a drastically different learning style.

Unlike traditional classifiers, LLMs are trained on wide semantic information learned in large scale pretraining, which allows them to understand complex and indirect clinical descriptions [41]. This capacity is further enhanced when used with Retrieval-Augmented Generation (RAG), which grounds predictions on external knowledge sources [5].

This combination enables the model to perform something more than shallow pattern matching, but a kind of contextual reasoning, which is particularly useful for low-frequency and ambiguous diagnoses.

> *LLMs push the task from frequency-based pattern recognition to knowledge-based reasoning, enabling better detection of rare and complex conditions.*

6.4 The Accuracy Trap: A Misleading Objective

One major finding of this study is the formation of what we can name the *accuracy trap*. Models trained to maximize overall accuracy seem highly effective based on conventional measures, yet miss clinically meaningful rare diseases.

This is because typical assessment measures implicitly presuppose balanced label distributions, which is not the case in ICD-10 coding. Accuracy improvements often come at the expense of an increased ability to discriminate between classes that are already dominant rather than an increased ability to discriminate between classes with therapeutic utility.

> *Optimizing for accuracy alone can lead to models that are statistically strong but clinically inadequate.*

This finding highlights the need to re-examine evaluation methodologies in clinical machine learning with more attention on rare-class performance and task-specific value.

6.5 Efficiency versus Clinical Nuance

Another key feature that is shown in the trials is the trade-off between computational efficiency and clinical expressiveness.

Classical approaches, like as XGBoost, are particularly efficient and scalable and thus suitable for high-throughput applications . Transformer models are a solid middle ground, offering decent performance at a fair computational cost. On the other hand, the resource needs of the LLM-RAG systems are much larger yet they perform better on difficult and unusual scenarios.

This trade-off is not merely a computational issue, but also has direct ramifications for deployment decisions.

The cost of computation must be balanced against the cost of clinical error.

In many clinical contexts, the consequences of missing a rare diagnosis may far outweigh the increased computational expense.

6.6 Calibration and Trustworthiness

In addition to accuracy, the trustworthiness of model outputs is a vital consideration in clinical settings. Our findings demonstrate that RAG-based models are better calibrated, meaning that their produced probabilities are more representative of the actual outcomes.

This enhanced calibration improves the confidence in model predictions, particularly in decision-support applications where confidence estimates play a role in guiding clinical decision-making [22]. In contrast, poorly calibrated models may be overconfident, which can lead to unsafe recommendations.

Also, the retrieval component of RAG provides interpretability for its predictions by providing supporting evidence, an important requirement for use in clinical workflows.

6.7 Implications for Clinical Deployment

Results show that there is no single best model architecture. The relevance of a model is determined by the given application context.

Transformer-based or classical models provide a good solution in high-volume administrative settings like billing and coding, where efficiency and consistency are key. Their high accuracy on frequent diagnoses guarantees reliable processing at large scale.

Conversely, in clinical decision support scenarios, especially those involving rare or complex conditions, LLM-RAG systems have clear advantages. They can incorporate external knowledge and reason in context, leading to improved sensitivity for diagnosis.

These observations naturally motivate hybrid approaches that take advantage of multiple paradigms. For example, a two-stage system might use a low-cost model for regular cases and pass on uncertain or infrequent cases to a more expressive LLM-based model.

6.8 Deployment Challenges

Despite their advantages, there are practical difficulties with LLM-based systems and they remain even with the advantages they offer. They can be computationally expensive and may not scale well in resource restricted contexts. In addition, the use of retrieval systems adds engineering complexity and reliance on external knowledge sources.

Nevertheless, their improved calibration and interpretability make them good candidates for human-in-the-loop systems, where transparency and reliability are vital.

6.9 Summary

The chapter shows that model selection is a set of trade-offs, not a single ideal solution. Transformer-based models are highly precise and efficient, classical models are simple and fast and LLM-RAG systems offer finer clinical nuance and reliability.

The main contribution of this work is the identification and systematic analysis of these tradeoffs. Most importantly, it shows the limitations of traditional evaluation criteria and the need to connect model objectives with clinical needs.

Chapter 7

Conclusion and Future Work

7.1 Summary of Findings

This thesis investigated the task of automatic ICD-10 coding from clinical text through a systematic comparison of classical machine learning models, deep neural architectures, transformer-based models and large language models (LLMs) with Retrieval-Augmented Generation (RAG).

The most important finding of this work is that traditional evaluation measures and especially accuracy are not appropriate for evaluating clinical NLP systems that work with highly imbalanced data. Transformer-based models such as BERT obtained the highest overall accuracy and Micro-F1 scores, but their performance was primarily boosted by frequent diagnostic categories. Consequently, these models were nearly useless for detecting rare but clinically important conditions.

LLM-based approaches, especially when augmented with retrieval, on the other hand, showed a behavior fundamentally different. Their overall accuracy was slightly lower, but they improved the detection of rare classes substantially, with up to 40$\times$ improvement in Rare-F1. This indicates that these models are better suited to the semantic and contextual complexity of clinical narratives.

We also show through calibration analysis that RAG-based models provide more reliable confidence estimates, indicating that they are not just more expressive, but more trustworthy in decision-support scenarios.

> *The primary contribution of this thesis is to identify a fundamental misalignment between traditional performance metrics and clinical utility in ICD-10 coding systems.*

7.2 Key Insights

The findings indicate three main insights. First, optimizing for aggregate accuracy leads to models that prioritize dominant patterns at the expense of rare conditions, leading to what can be described as a *accuracy trap*. Second , LLM-based systems introduce a paradigm shift from pattern recognition to knowledge-driven reasoning , which enables better handling of complex and low-frequency diagnoses . Third, model calibration is a crucial factor for clinical deployment, as reliable confidence estimates are necessary for safe and interpretable decision support.

Together, these insights emphasize the need to evaluate clinical NLP systems along multiple dimensions such as rare-class performance, robustness and reliability, in addition to traditional metrics.

7.3 Implications for Clinical Deployment

The results of this study suggest that different modeling paradigms are suitable for different operational contexts. Transformer-based models are a practical trade-off between accuracy and computational efficiency in high-throughput administrative environments such as billing and large-scale coding pipelines. Their ability to pick up on frequent patterns means they perform consistently on common diagnoses.

In the setting of clinical decision support, however, where it is important to identify rare or complex conditions, LLM-RAG systems have clear advantages. Their ability to leverage external knowledge and do context-aware reasoning helps in making nuanced and clinically meaningful predictions.

The results suggest that the choice of model should depend on the intended application rather than on a single performance metric. In practice, this can motivate the development of hybrid systems, combining efficient models for the routine cases with more expressive models for complex scenarios, dynamically.

7.4 Future Work

This work opens various intriguing future research possibilities.

One major trend is the creation of hybrid modeling architectures that incorporate many paradigms in one framework. These systems could harness the efficiency of transformer-based classifiers for high frequency situations and pass over rare or ambiguous cases to LLM-based models, thus balancing the trade-off between scalability and clinical sensitivity.

Another focus is on improving cost-efficient LLM pipelines. With methods like LoRA and quantization, large model fine-tuning has become much cheaper computationally, but there is still much to be done to enable scalable deployment in resource-constrained environments. including exploring adaptive inference algorithms and better retrieval methods.

Another key study direction is calibration and uncertainty modeling. Future study could investigate more advanced calibration techniques, uncertainty-aware prediction frameworks, and tighter integration with human-in-the-loop systems, where the model outputs are intended to enhance clinical judgment rather than replace it.

The next crucial step is to examine these ideas in real-world clinical settings. This includes integration with electronic health record systems, validation with clinicians and coding professionals, and assessment of usability, trust and workflow impact. These investigations are critical for bridging the gap between experimental performance and practical adoption.

7.5 Conclusions

This thesis demonstrates that the evaluation of ICD-10 coding systems cannot be restricted to standard definitions of accuracy. This work systematically investigates the trade-offs between predictive performance, clinical nuance, computational economy and dependability to provide a more holistic view on the design of clinical NLP systems.

In summary, the results suggest that future AI systems in healthcare should optimize for performance, interpretability and trustworthiness, rather than a single parameter, in order to promote safe and effective medical decision-making.

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